

A13 ENHANCED SUPPLY AIR | O (MAX : 3 PT)

Intent : Mitigate risks from indoor contamination and pollution sources, such as infectious disease particles and volatile organic compounds (VOC).

Summary : This WELL feature requires the projects to use supply air that is not recirculated or that is treated with carbon filters, media filters and/or Ultraviolet Germicidal Irradiation (UVGI).

Issue : Building materials, furnishings (e.g., carpets and furniture finishes), fabrics, cleaning products, personal care products, adhesives, solvents and air fresheners can all emit VOCs or semi-volatile organic compounds (SVOCs) into the indoor environment.^{1,2} VOCs include benzene, formaldehyde and other chemical compounds which, at high concentrations, can lead to the irritation of the nose and pharynx and have been associated with leukemia and nasopharyngeal cancer.^{3,4} Health effects can also include damage to the liver, kidneys and central nervous system.⁵ When humans exhale, they can release pathogens, such as COVID-19 or Influenza particles, which can remain suspended in the air for several hours or longer. These pathogens can be recirculated through ducts and HVAC systems of buildings, increasing the risk of transmission of disease indoors.⁶⁻⁸

Solutions : The simplest way to avoid recirculating contaminated air is to not recirculate it by supplying spaces with 100% outdoor air. Unfortunately, in certain climates this can result in high energy usage, although this can be mitigated through the use of heat recovery systems.⁹ In buildings where recirculated air is utilized, it can be treated to remove contaminants. Carbon is capable of filtering VOCs and ozone from the air that passes through.^{10,11} HEPA or near-HEPA filters can help remove virus particles, since the virus often travels as an attachment to a larger particle.^{12,13} UVGI systems can also be effective, both when irradiating the upper portion of the room or when installed in the air ducts, so long as they are powerful and/or the air speed is slow enough to provide a sufficient UV dose.^{13,14} Other technologies exist, but should be evaluated carefully. For example, more novel devices that release oxidants into rooms have less robust evidence for safety and efficacy outside of laboratory conditions.^{15,16} Finally, in-room air purifiers can be beneficial because the clean air is often provided within the breathing zone. For optimal performance, air filtration systems need to be maintained according to the manufacturer's instructions.

PART 1 IMPROVE SUPPLY AIR (MAX : 1 PT)

For All Spaces:

Option 1: 100% Outdoor air provisions

The following requirements are met:

- a. Each regularly occupied space is ventilated with outdoor air that has not recirculated from other rooms within the building.

OR

Option 2: Cleaning and purification devices

All regularly occupied spaces with recirculated air are treated with purification/cleaning system(s), either in the HVAC system or as a standalone device, which meet the following requirements:

- a. Use, at a minimum, two of the following air purification/cleaning technologies:
 1. Activated carbon filter.
 2. Media filter with an average removal efficiency of $\geq 75\%$ for particles 0.3-1 μm in size (e.g., MERV 14; F8; ePM1 75%).
 3. UVGI to treat the moving air OR upper-room UVGI.
- b. Comply with one of the following:
 1. Are validated under UL 2998 Zero Ozone Emissions Validation or Intertek Zero Ozone Verification.¹⁵
 2. Do not use electronic air cleaners.
- c. Are not designed to release ions, reactants, or other molecules into occupiable spaces to disinfect or clean the air.¹⁶
- d. Are sized appropriately to the room volume or area, based on manufacturer specifications.
- e. Are maintained according to the manufacturer's recommendations. Documentation of maintenance is submitted annually through the WELL digital platform.

PART 2 B PROVIDE CLEAN AIRFLOW RATES FOR CONTROL OF INFECTIOUS AEROSOLS (MAX : 2 PT)

For All Spaces:

Option 1: Regular Airflow Operation

The mechanical system provides all regularly occupied spaces with clean airflow rates for infection risk management as set in one of the following guidelines:

- a. ASHRAE 241-2023 Table 5-1 Equivalent Clean Airflow Rates (Section 6 - Clean Airflow Rate Equation or Appendix C - In-place Modeling Method).¹⁷
- b. 5 Equivalent Air Changes per hour (5 e/ACH) calculated in accordance with Appendix A1.¹⁸

OR

Option 2: Operation Under Infectious Disease Management Mode

The following requirements are met:

- a. The mechanical system is capable of providing all regularly occupied spaces with clean airflow rates for infection risk management as set in one of the following guidelines:
 1. ASHRAE 241-2023 Table 5-1 Equivalent Clean Airflow Rates (Section 6 - Clean Airflow Rate Equation or Appendix C - In-place Modeling Method).¹⁷
 2. 5 Equivalent Air Changes per hour (5 e/ACH) calculated in accordance with Appendix A1.¹⁸

- b. The project has an operational plan for Infection Risk Management Mode (IRMM) that includes the following:¹⁷
1. The conditions under which the IRMM will be activated.
 2. The engineering controls (e.g., changes in ventilation or air treatment) and non-engineering controls (e.g., changes in occupancy) that will be utilized to achieve the target clean airflow rates.
 3. The operational and maintenance procedures in place to implement target clean airflow rates within 48-hours of the decision to activate IRMM.
 4. An inventory of HVAC system consumables (e.g., filters, adsorption media, UV bulbs) that is updated when significant equipment modifications have been implemented or after replacement consumables have been utilized.
 5. A requirement that there is at least one set of replacement consumables available to install during IRMM.

Note :

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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